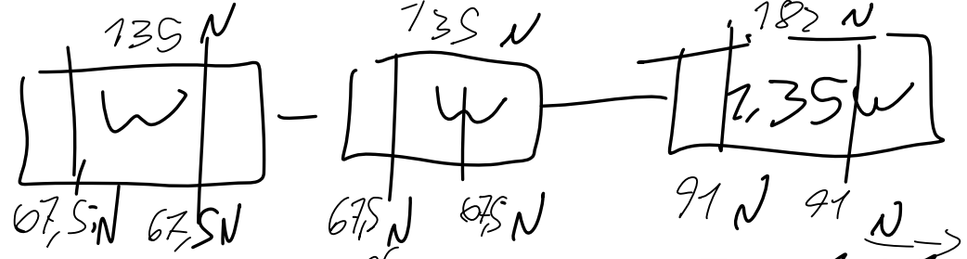
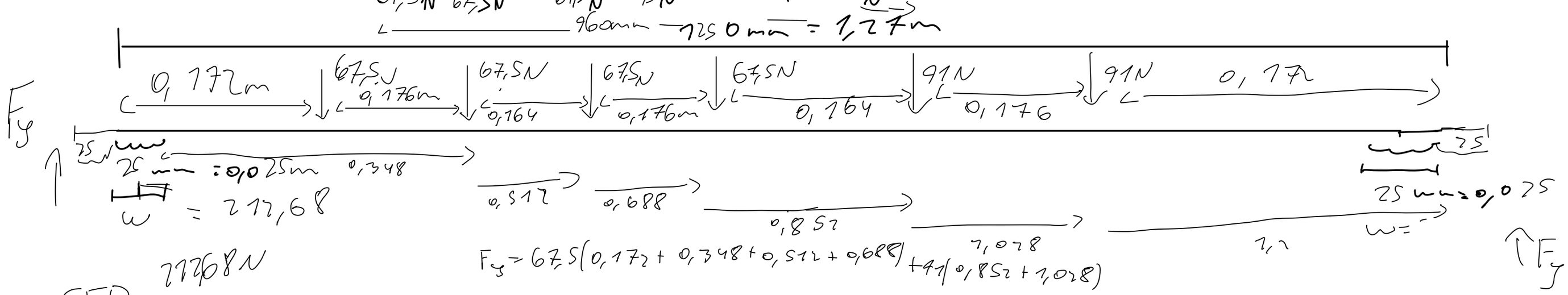
FBD



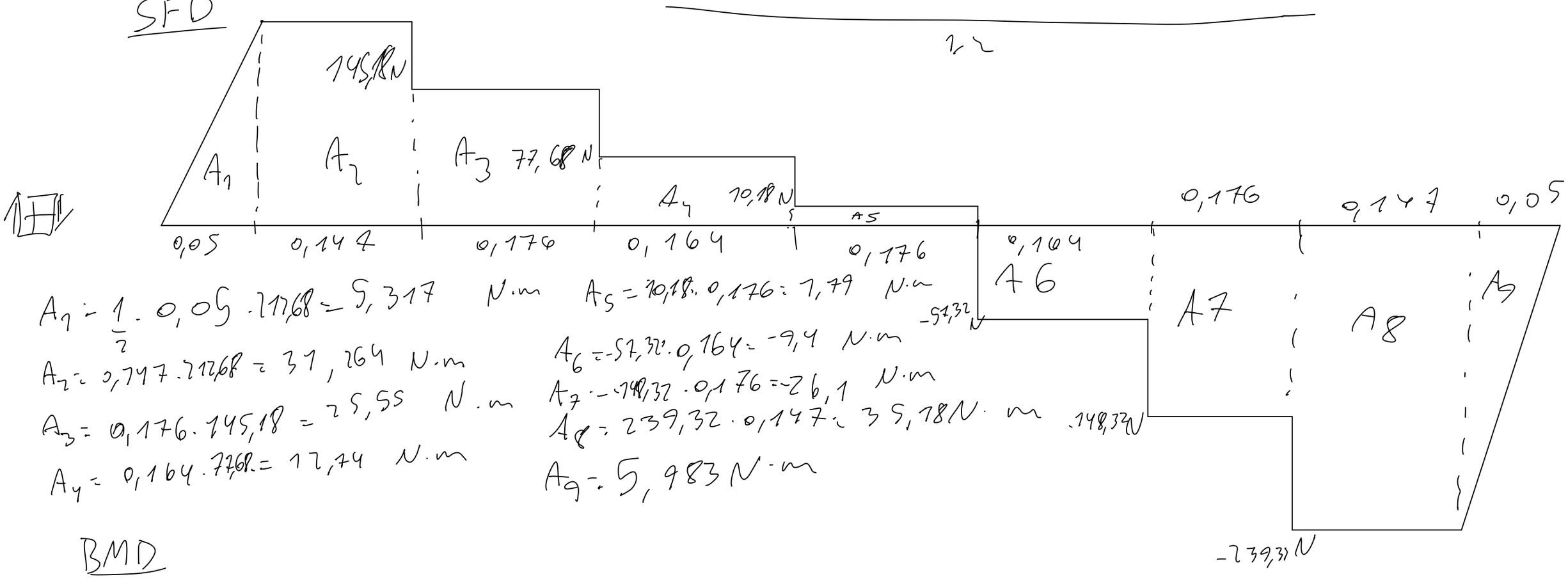
Total weight = 452 N

$F_y = 239{,}32\ N$



$F_y = \underbrace{67{,}5(0{,}172 + 0{,}348 + 0{,}512 + 0{,}688) + 91(0{,}852 + 1{,}028)}_{1{,}2}$

SFD



$A_1 = \frac{1}{2} \cdot 0{,}05 \cdot 272{,}68 = 5{,}317$ N·m

$A_2 = 0{,}147 \cdot 272{,}68 = 31{,}264$ N·m

$A_3 = 0{,}176 \cdot 145{,}18 = 25{,}55$ N·m

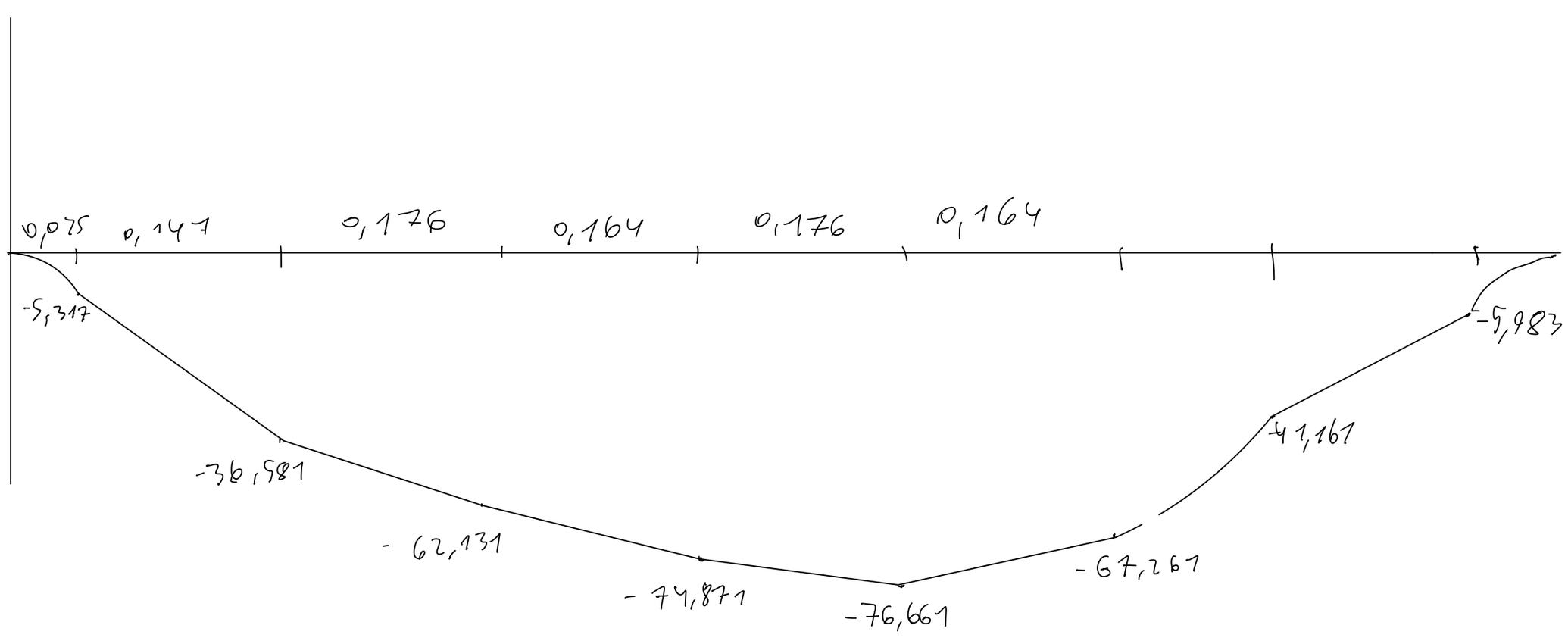
$A_4 = 0{,}164 \cdot 77{,}68 = 12{,}74$ N·m

$A_5 = 10{,}18 \cdot 0{,}176 = 1{,}79$ N·m

$A_6 = -57{,}32 \cdot 0{,}164 = -9{,}4$ N·m

$A_7 = -148{,}32 \cdot 0{,}176 = -26{,}1$ N·m

$A_8 = 239{,}32 \cdot 0{,}147 = 35{,}18$ N·m

$A_9 = 5{,}983$ N·m

BMD